Mini-Project Report (CS 2191N)
Data Exploration and Model Training for Credit Card Fraud Detection

Sravan Shanker
2025csb087

Dataset: https://www.kaggle.com/datasets/kartik2112/fraud-detection/
Github Repo: https://github.com/Sravcoding/mini-project-sem-3
Technology Used: Python, Numpy, Pandas, Matplotlib, Tqdm

Credit card fraud detection is complicated by severe class imbalance, where standard machine learning algorithms achieve misleadingly high accuracy by predicting the majority class while failing to catch fraudulent events. To address this using a 30,000 record dataset, we developed a systematic workflow. After cleaning irrelevant data columns, encoding string variables, and transforming features, we visualized and explored the data patterns, applied targeted resampling techniques to balance the class distribution, and trained various machine learning algorithms to build robust classification models.

The dataset comprises approximately 150,000 credit card transaction records, from which a subset of 30,000 instances was sampled due to hardware constraints. As typical for fraud detection tasks, the data is heavily imbalanced.

Training standard machine learning models on raw imbalanced data yields deceptively high accuracy by defaulting to the majority class while failing to identify critical fraudulent transactions. To address this, exploratory experiments evaluated various data resampling strategies. Additionally, irrelevant features were removed or transformed, and categorical string variables were encoded to preserve key information.

1. Data transformation

1. Feature Removal: Irrelevant columns—specifically first and last names (as credit card numbers uniquely identify users), transaction numbers, street addresses, ZIP codes, cities, states, and Unix timestamps—were dropped.
2. Geospatial Distance Calculation: Latitude and longitude coordinates were used to calculate the physical distance between the merchant and the cardholder at the time of each transaction.
3. Age Derivation: Date of birth was transformed into the user's current age.
4. Temporal Decomposition: Transaction timestamps were split into distinct cyclical components, specifically the hour of the day and the day of the week.
5. Numerical Retention: City population and transaction amounts were kept in their original numeric forms.
6. Categorical Encoding: Gender was processed using one-hot encoding.
7. Target Encoding for High-Cardinality Features: Merchant and job fraud rates were computed exclusively across the training set to identify high-risk merchants and occupations. These historical fraud rates were saved and mapped onto the test set,

capturing broader risk trends efficiently without resorting to complex high-cardinality encoding techniques.

```
train.head()
```

Python

	is_fraud	amt	city_pop	distance	hour	dow	gender_F	gender_M	merchant_fraud_rate	job_fraud_rate
0	0	4.97	3495	78.597568	0	2	1.0	0.0	0.0	0.0
1	0	107.23	149	30.212176	0	2	1.0	0.0	0.0	0.0
2	0	220.11	4154	108.206083	0	2	0.0	1.0	0.0	0.0
3	0	45.00	1939	95.673231	0	2	0.0	1.0	0.0	0.0
4	0	41.96	99	77.556744	0	2	0.0	1.0	0.0	0.0

2. Data Exploration and Visualization

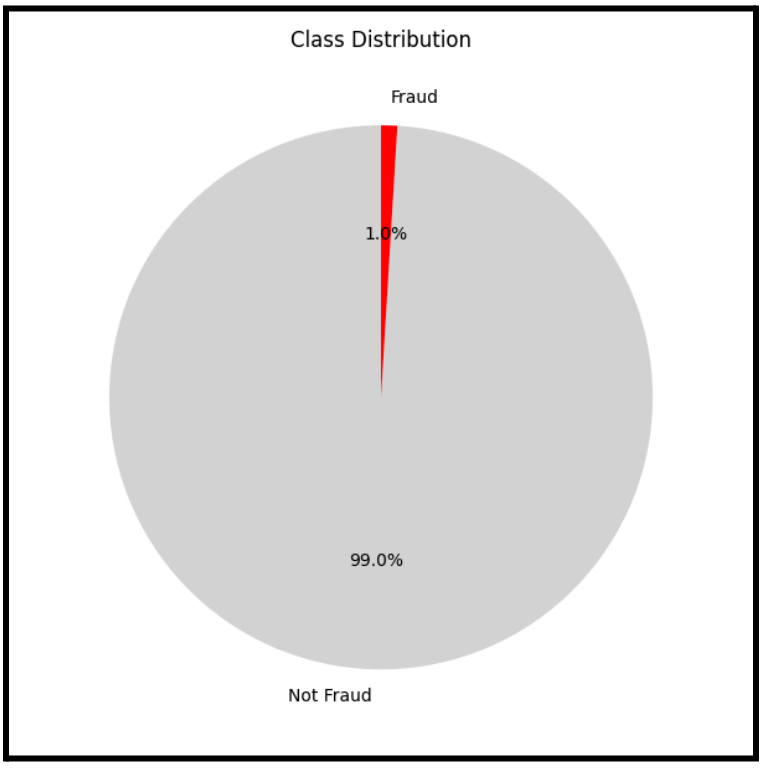
Global fraud is 0.96%. All outliers in terms of fraud are compared to this universal value, as marked by red lines in graphs wherever possible.

A. Data Overview and Class Imbalance

Payment fraud is rare but costly. Across the entire dataset, transactions show an extreme class imbalance:

- Legitimate Transactions: 99.0% (the vast majority of activity)
- Fraudulent Transactions: 1.0% (the baseline risk rate)

Because fraud represents only 1 in every 100 transactions, detection systems must focus on specific risk indicators rather than overall transaction volume.

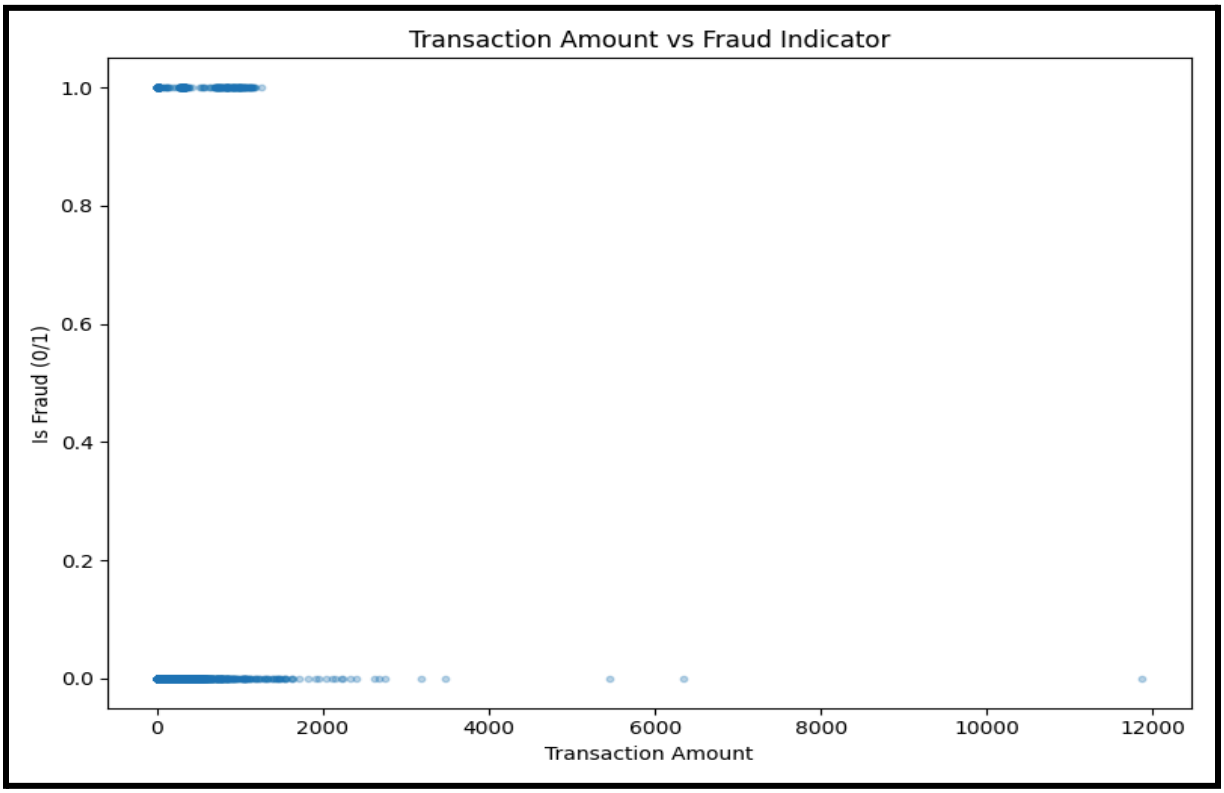


B. Transaction Amount Limits

Fraudulent transactions do not involve huge dollar amounts. The scatter plot of transaction amounts shows a clear monetary pattern:

- Fraud Ceiling: All fraudulent transactions occur below $1,400.
- High-Value Purchases: Purchases above 1,400 (extending up to **11,900**) are 100% legitimate.

Fraudsters deliberately keep transactions small to stay under automated alert thresholds and avoid quick detection.



C. Risk by Merchant Category

Merchant categories show varying levels of risk compared to the overall 1.0% baseline:

High-Risk Categories

- Online Shopping (shopping_net): 2.87% fraud rate (nearly 3x the baseline).
- Physical Grocery (grocery_pos): 2.72% fraud rate.
- Miscellaneous Online (misc_net): 2.38% fraud rate.

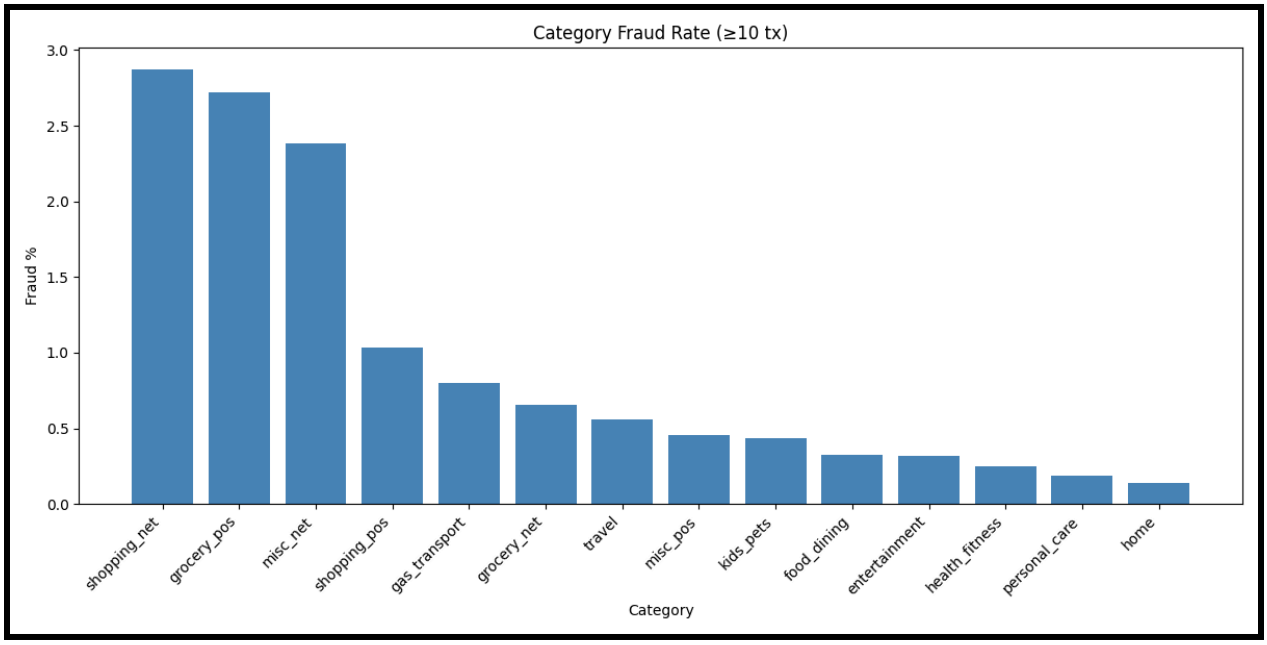
Moderate-Risk Categories

- Physical Shopping (shopping_pos): 1.03%
- Gas & Transport (gas_transport): 0.80%

- Online Grocery (grocery_net): 0.65%
- Travel (travel): 0.56%

Low-Risk Categories (<0.5%)

Categories like misc_pos (0.45%), kids_pets (0.43%), food_dining (0.32%), entertainment (0.31%), health_fitness (0.25%), personal_care (0.18%), and home (0.14%) carry very low risk.



D. Time and Schedule Patterns

Fraud activity follows strong hourly, daily, and weekly cycles.

Hourly Trends: Late-Night Risk

- Late-Night Peak (10 PM – 3 AM): Fraud spikes significantly at night, reaching 4.58% at 10 PM and 4.36% at 11 PM. Elevated risk continues from midnight to 3 AM (2.22% to 2.72%).
- Daytime Lows (4 AM – 9 PM): Fraud rates drop below 0.5% during normal waking hours, falling below 0.10% between 4–5 AM and 5–7 PM.

Daily and Weekly Trends

- Daily Peaks: Across the 18-day tracking period (Jan 1–18, 2019), fraud peaked on January 9 (1.98%) and stayed high between January 12–16 (1.44%–1.68%).
- Day of Week: Wednesday (1.45%) and Saturday (1.18%) show the highest risk. Sunday has the lowest risk (0.77%).

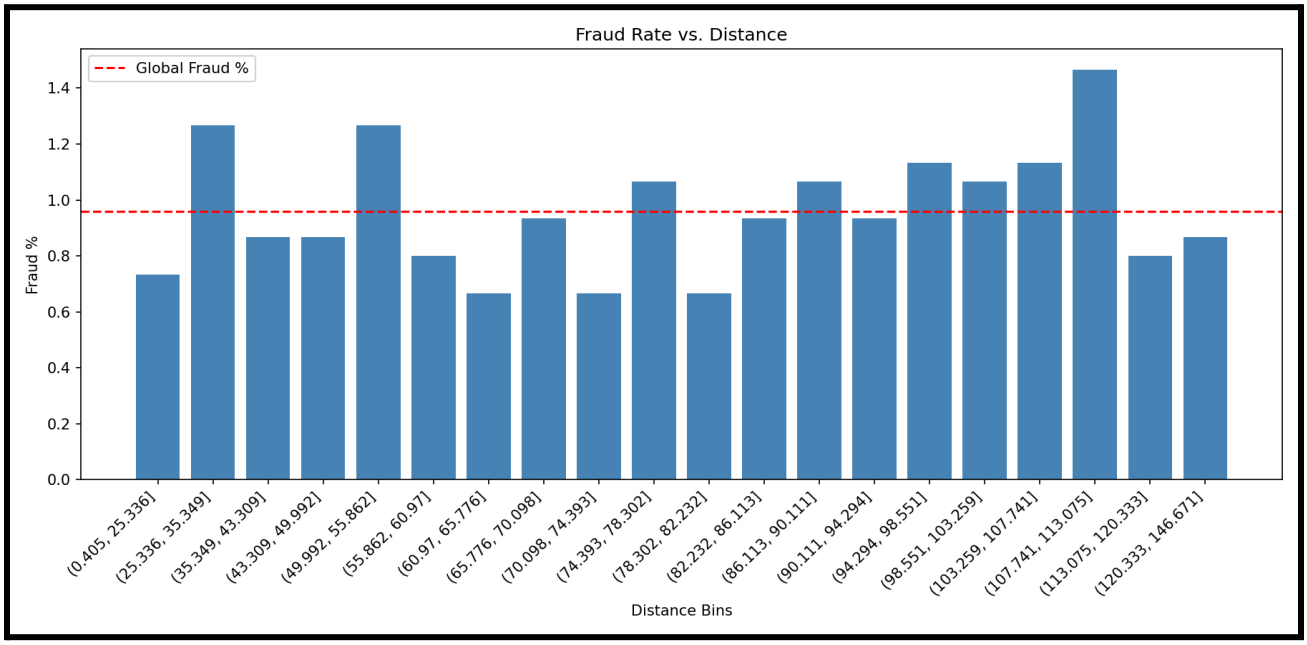


E. Location and Distance Effects

Transaction distance (between cardholder residence and purchase location) impacts fraud risk non-linearly against the 0.96% distance baseline:

- Highest Risk Distance Bin: (107.7 to 113.1 units) reaches 1.46% fraud rate (+52% above baseline).

- Secondary Risk Peaks: (25.3 to 35.3 units) and (50.0 to 55.9 units) both reach 1.26%.
- Local Transactions: (0.4 to 25.3 units) show below-average risk at 0.73%.
- Lowest Risk Bins: Certain mid-to-long distance bins drop to 0.66%.



F. Compromised Professions and Merchants

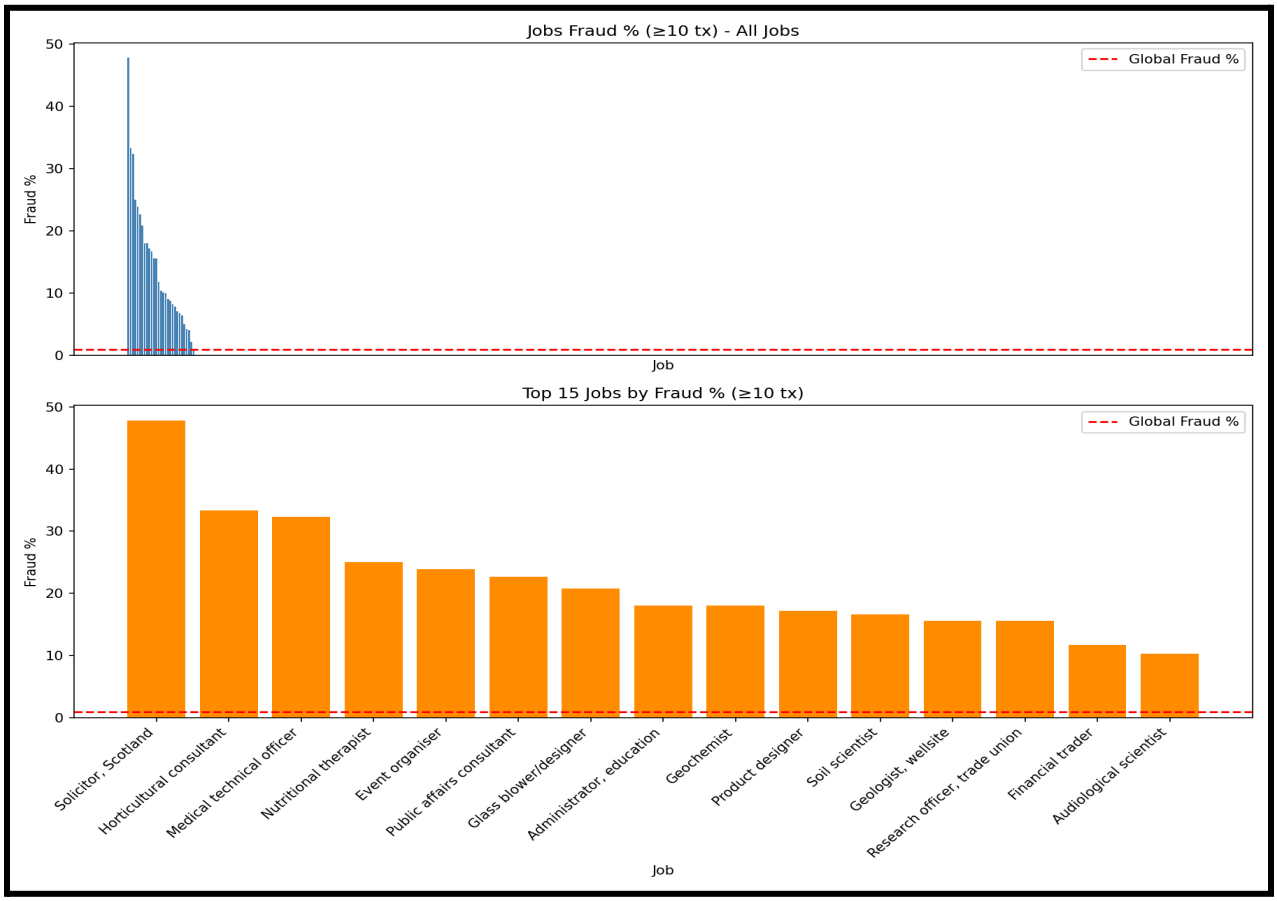
Fraud is heavily concentrated among specific cardholder job titles and vendor accounts (minimum 10 transactions).

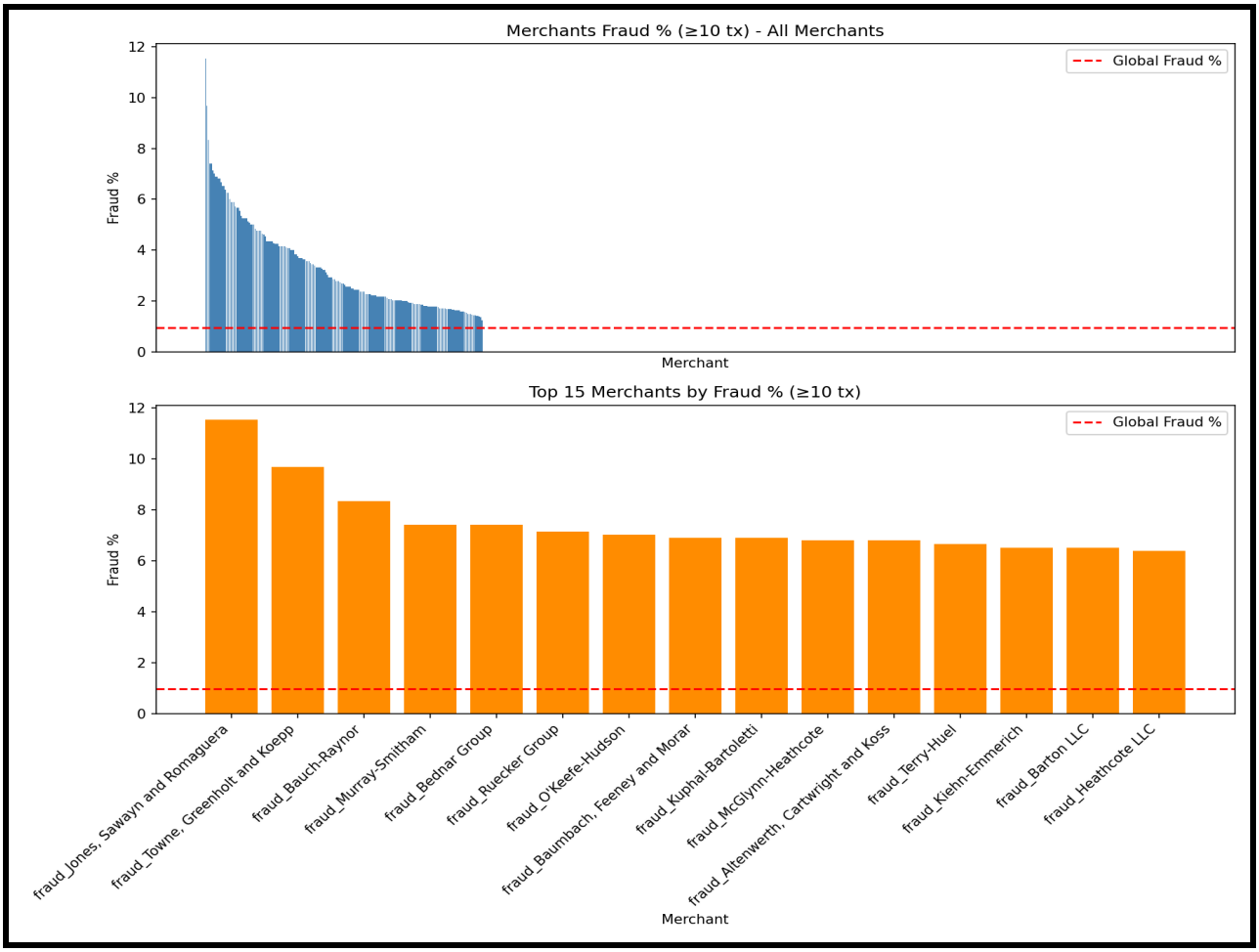
Top High-Risk Job Titles

1. Solicitor, Scotland: 47.6% fraud rate
2. Horticultural consultant: 33.3%
3. Medical technical officer: 32.3%
4. Nutritional therapist: 25.0%
5. Event organiser: 23.8%
6. Public affairs consultant: 22.6%
7. Glass blower/designer: 20.7%

Top High-Risk Merchants

8. fraud_Jones, Sawayn and Romaguera: 11.5% fraud rate (11.5x the baseline)
9. fraud_Towne, Greenholt and Koepp: 9.7%
10. fraud_Bauch-Raynor: 8.3%
11. fraud_Murray-Smitham & fraud_Bednar Group: 7.4%
12. fraud_Ruecker Group: 7.1%


Jobs Fraud % (≥10 tx) - All Jobs
Global Fraud %
Fraud %
Job
Top 15 Jobs by Fraud % (≥10 tx)
Solicitor, Scotland
Horticultural consultant
Medical technical officer
Nutritional therapist
Event organiser
Public affairs consultant
Glass blower/designer
Administrator, education
Geochemist
Product designer
Soil scientist
Geologist, wellsite
Research officer, trade union
Financial trader
Audiological scientist

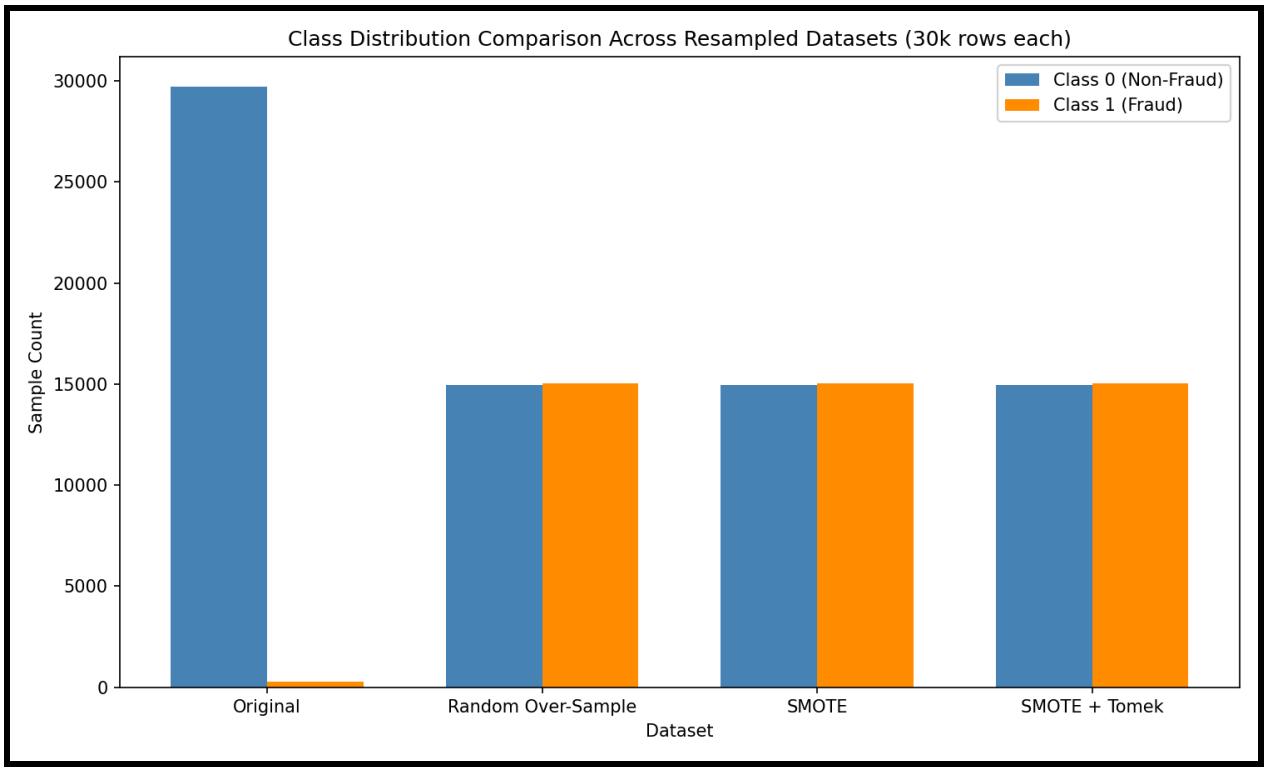

Merchants Fraud % (≥10 tx) - All Merchants
Global Fraud %
Fraud %
Merchant
Top 15 Merchants by Fraud % (≥10 tx)
fraud_Jones, Sawayn and Romaguera
fraud_Towne, Greenholt and Koepp
fraud_Bauch-Raynor
fraud_Murray-Smitham
fraud_Bednar Group
fraud_Ruecker Group
fraud_O'Keefe-Hudson
fraud_Baumbach, Feeney and Morar
fraud_Kuphal-Bartoletti
fraud_McGlynn-Heathcote
fraud_Altenwerth, Cartwright and Koss
fraud_Terry-Huel
fraud_Kiehn-Emmerich
fraud_Barton LLC
fraud_Heathcote LLC

3. Comparative Analysis of Resampling Frameworks

The class distribution comparison visualization examines four distinct dataset states across uniform 30,000-row evaluation samples. The baseline unadjusted dataset contains approximately 29,700 non-fraudulent instances and fewer than 300 fraudulent instances.

Resampling Strategy	*Class 0 (Non-Fraud) Count*	*Class 1 (Fraud) Count*	*Target Balance Ratio*	*Operational Characteristics & Modeling Impact*
Original Dataset	~29,700	~300	99.0% / 1.0%	Severe class imbalance; high risk of model bias toward majority non-fraud class.
Random Over-Sampling	15,000	15,000	50.0% / 50.0%	Duplicates minority fraud instances; balances class weight but risks overfitting.
SMOTE	15,000	15,000	50.0% / 50.0%	Synthesizes novel minority samples along k-NN vectors; expands decision boundary.
SMOTE + Tomek Links	~14,970	15,000	~49.95% / 50.05%	Synthesizes minority instances while removing ambiguous border-pair noise (Tomek links).

As shown in the comparative resampling metrics, both Random Over-Sampling and standard SMOTE achieve a perfect 1:1 balance (15,000 samples per class). However, the hybrid SMOTE + Tomek Links method introduces data cleaning by removing Tomek links—pairs of opposing class instances that sit in close proximity. This slightly reduces the majority class count (~14,970) while preserving 15,000 synthetic minority instances, yielding cleaner spatial boundaries for classifier optimization.



4. Model Training and Optimization

For credit card fraud, it is quite easy to get high accuracy by simply predicting the majority class. Hence we must also pay close attention to Recall (TP/TP+FN) to ensure a robust system and to minimise False Negatives.
We have swept over parameter values over each sampling technique database: as is, random oversampling, smote and smote + tomek.

We have used 5 different models:
Note: During training we also used a Decision Tree model, but could not get satisfactory results and as such we are excluding it from the report.

1. Regularized Logistic Regression:

Parameter:	
C	0.1
Threshold	0.3
Class Weight	1 : 50.0

Learning Rate	0.01
Epoch	5000
Dataset	No Resampling
Best Result:	
Accuracy	90.38%
Recall	77.77%

2. Neural Network with L2 Loss:

Parameter:	
Layer Shape	9 > 64 > 32 > 2
Threshold	0.7
Class Weight	1 : 1.5
Learning Rate	0.1
Epoch	1000
Dataset	Random Oversample
Best Result:	
Accuracy	86.82%
Recall	83.33%

3. Support Vector Machine (SVM):

Parameter:	
Kernel	Gaussian
Sigma for kernel	2.0
Threshold	-.0.1
C	0.5
Dataset	No Resampling

Best Result:	
Accuracy	89.92%
Recall	83.33%

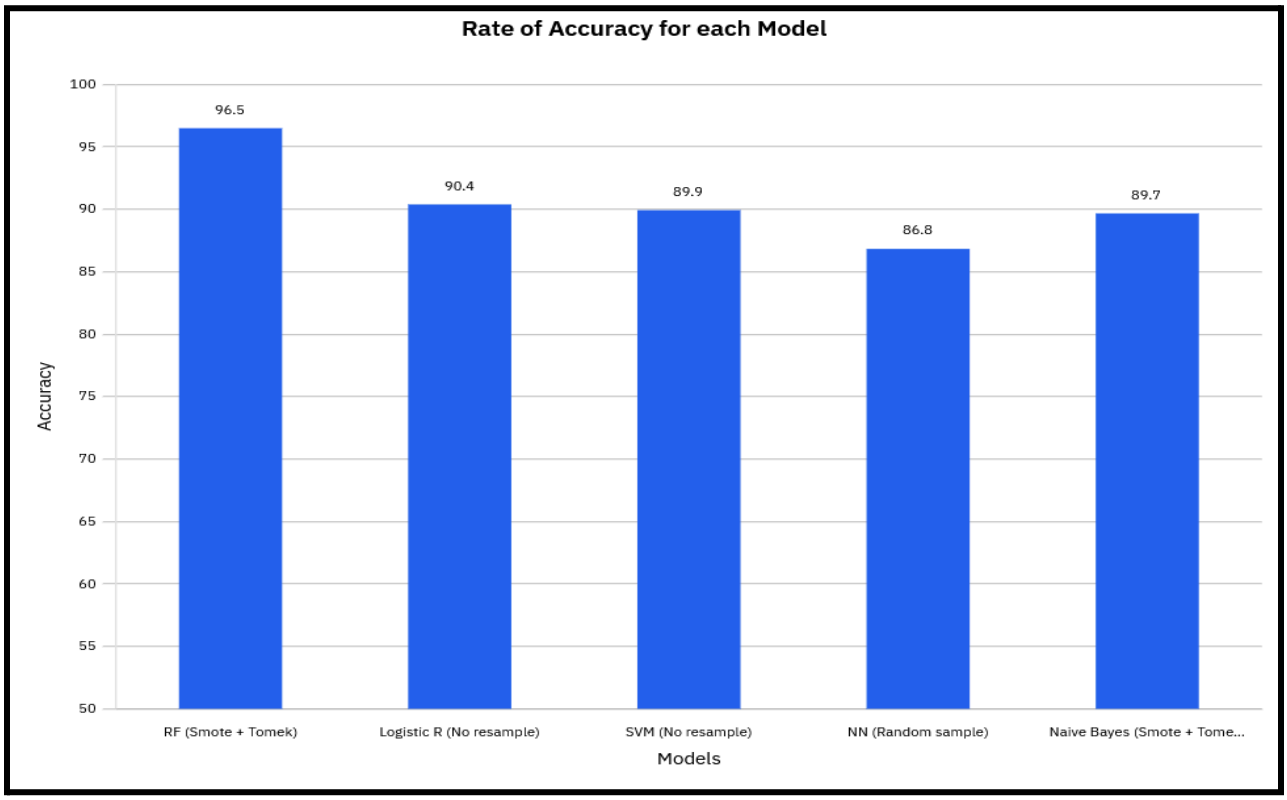
4. RandomForest:

Parameter:	
Threshold	0.2
Max Depth	10
N estimators	30
Min split	5
Dataset	Smote + Tomek
Best Result:	
Accuracy	96.48%
Recall	83.33%

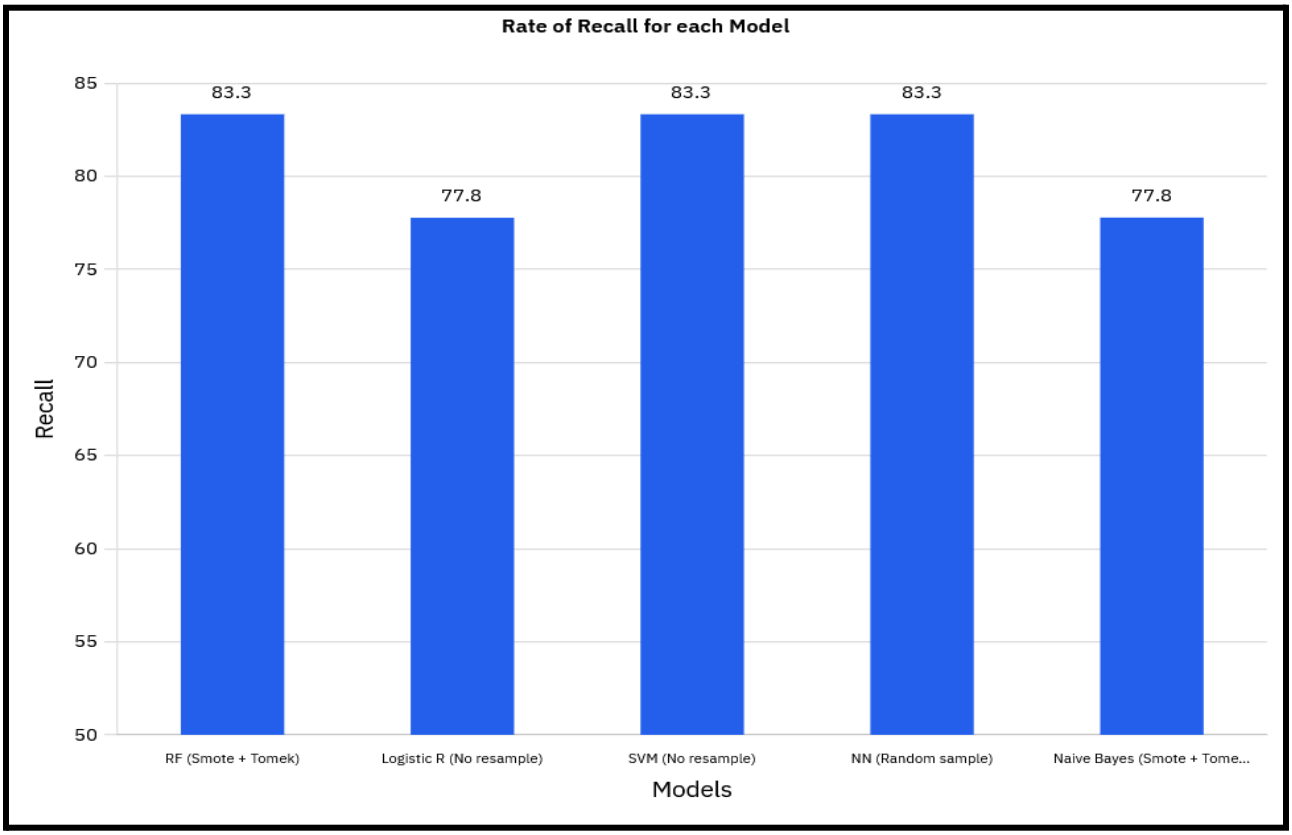
5. Gaussian Naive Bayes:

Parameter:	
Threshold	0.7
Class Prior	0.05 : 0.95
Dataset	Smote + Tomek
Best Result:	
Accuracy	89.66%
Recall	77.78%

Accuracy Graph:



Recall Graph:



5. Conclusion

In summary, this project successfully addressed the severe class imbalance inherent in credit card fraud detection by combining rigorous feature engineering—such as geospatial distance calculation, temporal decomposition, and target-encoded risk rates—with advanced resampling frameworks like SMOTE + Tomek.

Through systematic parameter and threshold sweeping across multiple classifiers, the **Random Forest model** trained on **SMOTE + Tomek data** emerged as the most robust approach, achieving an impressive **96.48% accuracy** and **83.33% recall**.

Ultimately, this work demonstrates that balancing class distributions alongside careful decision-threshold tuning is vital for building effective fraud detection systems that successfully catch critical financial crimes while maintaining high overall reliability.

6. Bibliography

A. SVM:
https://www.marenglenbiba.net/dm/cortes_vapnik95.pdf

https://www.youtube.com/watch?v=lDwow4aOrtg&list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU&index=6&pp=iAQB

B. Neural Network:
https://www.youtube.com/watch?v=MfIjxPh6Pys

https://www.geeksforgeeks.org/machine-learning/what-is-cross-entropy-loss-function/

C. Decision Tree and Random Forest:
https://www.youtube.com/watch?v=wr9gUr-eWdA&list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU

https://medium.com/@nielspace/building-a-decision-tree-regression-model-from-scratch-part-1-784d349b86af

D. Dataset Balancing:
https://www.researchgate.net/publication/398128653_Using_SMOTE_and_TOMEK_Link_Sampling_Techniques_to_Address_Imbalanced_Data_Challenges_in_the_Machine_Learning_models

E. Naive Bayes:
https://www.youtube.com/watch?v=nt63k3bfXS0&list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU&index=5